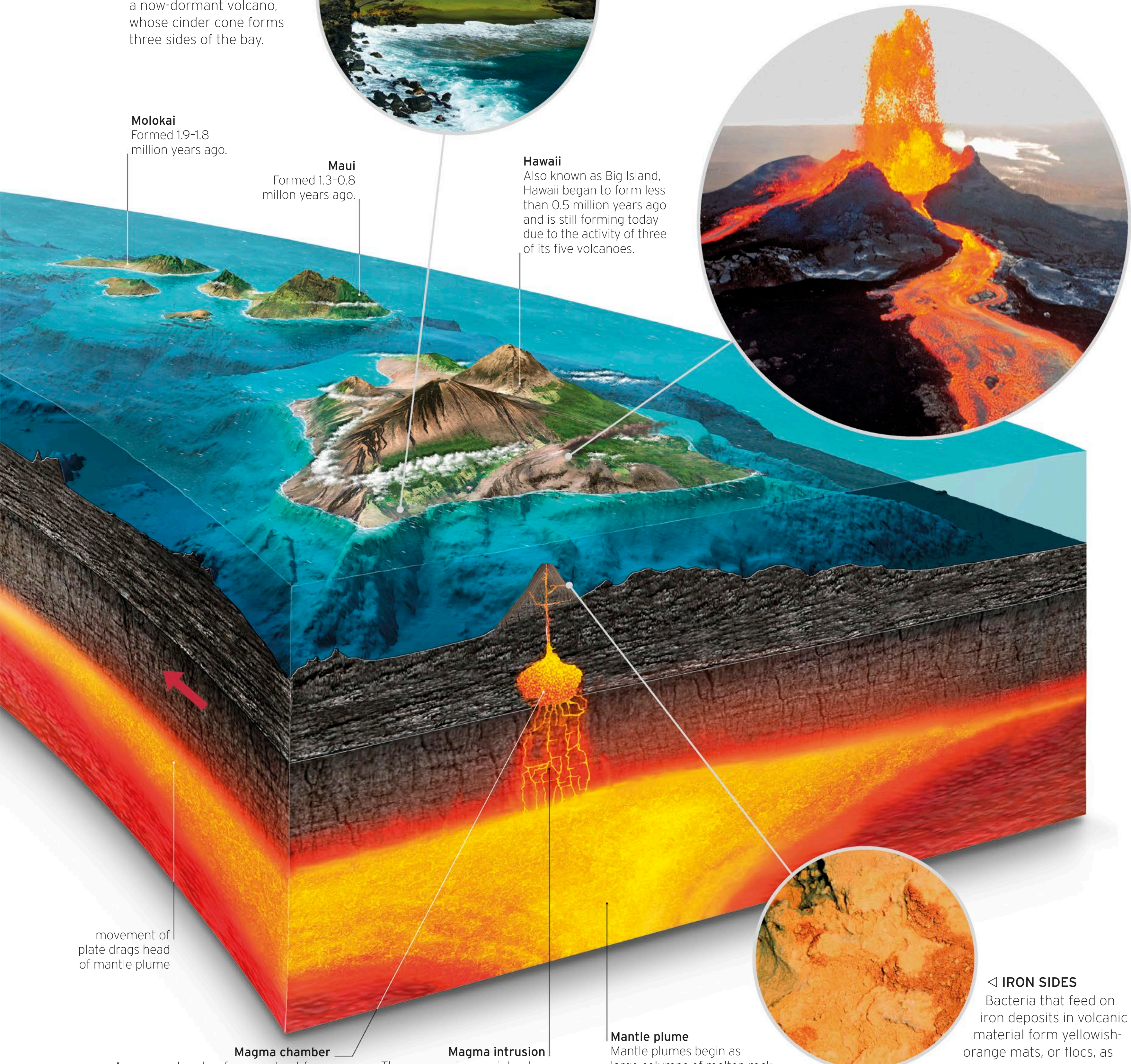
▷ **GREEN SAND**
One of only a handful of green sand beaches in the world, Papakolea borders Mahana Bay, on the southern tip of Hawaii. The sand gets its color from olivine, a mineral released by eruptions from a now-dormant volcano, whose cinder cone forms three sides of the bay.

▽ **LAND BUILDER**
One of the world's most active volcanoes, Kilauea's eruptions appear violent, but are tame compared to more explosive continental volcanoes. Its basaltic lava flows build layer upon layer of what will one day be fertile soil.



Molokai
Formed 1.9–1.8 million years ago.

Maui
Formed 1.3–0.8 millon years ago.

Hawaii
Also known as Big Island, Hawaii began to form less than 0.5 million years ago and is still forming today due to the activity of three of its five volcanoes.

movement of plate drags head of mantle plume

Magma chamber
A magma chamber forms as heat from the mantle plume melts part of the lithosphere. As the plate moves across a magma chamber, the molten rock pushes upward in weaker areas, creating a volcano that punches through the plate.

Magma intrusion
The magma rises, or intrudes, into lower-density rock, creating dikes. Eventually enough dikes are created to allow magma to collect in a chamber.

Mantle plume
Mantle plumes begin as large columns of molten rock (magma), which form deep within Earth's interior and rise through the mantle. Upon reaching the base of the lithosphere, the magma spreads, forming a cap or plume.

◁ **IRON SIDES**
Bacteria that feed on iron deposits in volcanic material form yellowish-orange mats, or flocs, as they rust (oxidize) the metal. Many such flocs have been found on the submerged flanks of the Loihi seamount, Hawaii's youngest submarine volcano.